

Xtra –

Note on masters theorem

–Continuous Generalized Master Theorem

$$T(n) = aT\left(\frac{n}{b}\right) + c \times f(n)$$

$$\text{or, } 2T\left(\frac{N}{2}\right) + C \times N^0$$

Step 1 : Extract the variables:

$$a = 2$$

$$b = 2$$

$$f(N) = N^0$$

Step 2 Calculate $n^{\log_b a}$:

$$\therefore N^{\log_b a} = N^{\log_2 2} = N^1 = N$$

Step 3: Determine ε :

- **Exponent of $N^{\log_b a}$ is 1.**
- **Exponent of $f(n) = N^0 = 0$.**
- **Difference (ε) is : $1 - 0 = 1$.**
- **Therefore $\varepsilon = 1$.**

$\varepsilon = \text{The exponential gap} = \text{Exponent of } N^{\log_b a} - \text{Exponent of } f(n)$

$1 - 0 = 1$ and this is not universal for three cases .what are the cases :

Cases

1. If there exists a constant $\varepsilon > 0$ such that

$$f(n) = O(n^{\log_b a - \varepsilon}) \Rightarrow n^{\log_b a - \varepsilon} < n^{\log_b a} \text{ then } T(n) = \Theta(n^{\log_b a}).$$

2. If there exists a constant $k \geq 0$ such that

$$f(n) = \Theta(n^{\log_b a} \log^k n), \text{ then } T(n) = \Theta(n^{\log_b a} \log^{k+1} n).$$

3. If there exists a constant $\varepsilon > 0$ such that

$f(n) = \Omega(n^{\log_b a + \varepsilon})$, and if $f(n)$ additionally satisfies the regularity condition

$$af\left(\frac{n}{b}\right) \leq cf(n) \text{ for some constant } c < 1 \text{ and}$$

all sufficiently large n , then $T(n) = \Theta(f(n))$.

Case 1

$$f(n) = O(n^{\log_b a - \varepsilon})$$

Here , $\log_b a - \varepsilon$, this means $f(n)$ grows smaller .

" $f(n)$ grows more slowly than $n^{\log_b a}$."

or

" $f(n)$ has a lower asymptotic growth rate than $n^{\log_b a}$."

***∴ Property 0 :** Exponent Gap $\varepsilon :=$ Exponent of $N^{\log_b a}$ – Exponent of $f(n)$*

Property 1: Exponent of $f(n) < \text{Exponent of } N^{\log_b a}$

According to our example:

$$2T\left(\frac{N}{2}\right) + C \times N^0$$

$\therefore$ For Case 1, $\varepsilon > 0$ and exponential gap is : 1, there for range of ε is: $0 < \varepsilon \leq 1$

ε , is true for 0.1, 0.2 1

Property 2: $n^{\log_b a - \varepsilon} < n^{\log_b a}$

Property 3: $\log_b a - \varepsilon < \log_b a$

$$n^{\log_2 2 - 0.1} < n^{\log_2 2} \text{ or } n^{1-0.1} < n^1 \text{ [satisfies]}$$

$$n^{\log_2 2 - 0.2} < n^{\log_2 2} \text{ or } n^{1-0.2} < n^1 \text{ [satisfies]}$$

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$$n^{\log_2 2 - 1} < n^{\log_2 2} \text{ or } n^{1-1} < n^1 \text{ [satisfies]}$$

We generally takes the largest gap, as it easier approach and also mathematically correct.

$$i.e., n^{\log_2 2 - 1} < n^{\log_2 2} \text{ or } n^{1-1} < n^1 \text{ [satisfies]}$$

Also,

$$f(n) = O(n^{\log_b a - \varepsilon}) = O\left(\frac{n^{\log_b a}}{n^\varepsilon}\right)$$

$$= O\left(\frac{n^{\log_b a}}{n^\varepsilon}\right)$$

$$O\left(\frac{n^{\log_2 2}}{n^{0.1}}\right) \approx O\left(\frac{n^{\log_2 2}}{n^{0.2}}\right) \approx \dots \approx O\left(\frac{n^{\log_2 2}}{n^1}\right) = O(1).$$

Case 2

2. If there exists a constant $k \geq 0$ such that

$$f(n) = \Theta(n^{\log_b a} \log^k n), \text{ then } T(n) = \Theta(n^{\log_b a} \log^{k+1} n)$$

But this is only possible, when ?

$$\text{Exponent Gap } \varepsilon: \text{Exponent of } N^{\log_b a} - \text{Exponent of } f(n) = 0$$

But our exponent gap is :

$$\varepsilon = \text{The exponential gap} = \text{Exponent of } N^{\log_b a} - \text{Exponent of } f(n) = (1 - 0) = 1$$

Case 2 requires the exponent gap to be

$$0$$

Also this means : $n^{\log_b a - \varepsilon} = n^{\log_b a}$ i.e. $n^{\log_b a} = n^{\log_b a}$ as, $\varepsilon = 0$.

because the polynomial exponents must be equal.

But here,

$$1 \neq 0$$

Therefore,

$$f(N) = \Theta(N^0) \neq \Theta(N^{\log_b a} \log^k N).$$

$$\text{or, } f(N) = \Theta(N^0) \neq \Theta(N^1 \log^k N).$$

Hence,

Master Theorem Case 2 is not satisfied.

Also if we continue :

We got $\varepsilon > 0$ i.e. $\varepsilon = 1$, for Case 2 it will be required:

$$f(n) = \Theta(n^{\log_b a} \log^k n)$$

$$\begin{aligned} \text{even if } k = 0, f(n) &= \Theta(n^{\log_2 2} \log^0 n) \Rightarrow \Theta(n^{\log_2 2} \times (\log n)^0) = \Theta(n^{\log_2 2} \times 1) \\ &= \Theta(n) [\log_a a = 1] \end{aligned}$$

But our $f(n) = C \times N^0$ or, $\Theta(1)$.

$$f(n) = \Theta(n) \neq f(n) = \Theta(1)$$

not satisfied.

Property 1: Exponent of $f(n) = \text{Exponent of } N^{\log_b a}$

Property 2: Exponent Gap ε : Exponent of $N^{\log_b a} - \text{Exponent of } f(n) = 0$

Property 3: $n^{\log_b a - \varepsilon} = n^{\log_b a}$ i.e. $n^{\log_b a} = n^{\log_b a}$, as $\varepsilon = 0$.

Property 4: $\log_b a - \varepsilon = \log_b a \Rightarrow \log_b a = \log_b a$, as $\varepsilon = 0$.

Case 3

If there exists a constant $\varepsilon > 0$ such that $f(n) = \Omega(n^{\log_b a + \varepsilon})$, and if $f(n)$ additionally satisfies the regularity condition $af\left(\frac{n}{b}\right) \leq cf(n)$ for some constant $c < 1$ and all sufficiently large n , then $T(n) = \Theta(f(n))$.

Here ,

$$f(n) = \Omega(n^{\log_b a + \varepsilon})$$

Here , $\log_b a + \varepsilon$, this means $f(n)$ grows faster .

" $f(n)$ grows more faster than $n^{\log_b a}$."

or

" $f(n)$ has a faster asymptotic growth rate than $n^{\log_b a}$."

$\therefore$ Exponent Gap ε : Exponent of $f(n)$ – Exponent of $N^{\log_b a}$

Property 0 :

For Case 3, there is no fixed range for exponential gap for ε , just here written If there exists a constant $\varepsilon > 0$, doesn't signifies a range for ε unlike for case 1 has :

$$0 < \varepsilon \leq \left(\text{Exponent of } N^{\log_b a} - \text{Exponent of } f(n) \right)$$

Hence we have to search if there exists a constant $\varepsilon > 0$.

Property 1: Exponent of $N^{\log_b a} < \text{Exponent of } f(n)$

Property 2: Exponent Gap $\varepsilon := \text{Exponent of } f(n) - \text{Exponent of } N^{\log_b a}, \varepsilon > 0$

Property 3: $n^{\log_b a} < n^{\log_b a + \varepsilon}$

Property 4: As $\varepsilon > 0$, hence, $\log_b a + \varepsilon > \log_b a$

Exponent of $N^{\log_b a}$ is 1 $\nless$ Exponent of $f(n) = 0$ [**Property 1 fails**]

Exponent Gap $\varepsilon := \text{Exponent of } f(n) - \text{Exponent of } N^{\log_b a} = 0 - 1 = -1$

[Negative] and $\varepsilon \nless 0$ [**Property 2 fails**]

$n^{\log_b a} < n^{\log_b a + \varepsilon} = n^{\log_2 2} < n^{\log_2 2 + (-1)} = n^1 \nless n^0$ [**Property 3 fails**]

$\log_b a + \varepsilon > \log_b a = \log_2 2 + (-1) > \log_2 2 = 0 \nless 1$ [**Property 4 fails**]

Hence Case 3 cannot be satisfied [This is the proper way]

**What earlier we did is? Taking $\varepsilon = 1$ as Universal test case
for case 1, 2 and case 3, which is ok but it is not a universal rule,
for case 1, 2, 3 as we have observed above, hence, we must avoid.**

Additional for Case 3

And for case 3 , further , we know :

Property 0 :

*For Case 3, there is no fixed range for exponential gap for ϵ ,
just here written If there exists a constant $\epsilon > 0$,
doesn't signifies a range for ϵ unlike for case 1 has :*

$$0 < \epsilon \leq \left(\text{Exponent of } N^{\log_b a} - \text{Exponent of } f(n) \right)$$

But , in general for case 3:

ϵ is the range between $\epsilon :=$

$$0 < \epsilon \leq \left(\text{Exponent of } f(n) - \text{Exponent of } N^{\log_b a} \right)$$

i. e. if $\epsilon_{\max} = 1$, it must also satisfy , 0.5, 0.2, 0.01 ... like case 1.

*And generally in maths we continue with $\epsilon_{\max}$ avoiding the range : $0 < \epsilon \leq \epsilon_{\max}$
, where $\epsilon_{\max} = g$ i. e. exponential gap.*

Summary

- ***Case 1: If the exponent gap is $g > 0$, then***

$$\boxed{0 < \epsilon \leq g.}$$

where $g = \text{Exponent of } N^{\log_b a} - \text{Exponent of } f(n)$

- ***Case 2: Exponent gap = 0, so ϵ is not used.***

- *Case 3: If the exponent gap is $g > 0$, then*

$$\boxed{0 < \varepsilon \leq g.}$$

where $g = \text{Exponent of } f(n) - \text{Exponent of } N^{\log_b a}$
